

Oluwamayowa Kalejaiye

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PUBLICATIONS

- Kalejaiye, O. (2026). Failures: Deterministic Failure-Mode Guardrails Make Coding Agents
- Build Resilient Systems. Zenodo, 15 pp. <https://doi.org/10.5281/zenodo.22966362>
- Open-source artifact: github.com/mayowa-kalejaiye/Failures

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, SQL | Backend: FastAPI, Django, Next.js, Node.js
- **Data & Infra:** PostgreSQL, MongoDB, Firebase, REST, WebSockets, Git, Vercel
- **AI:** LLM applications (OpenAI, Claude, Gemini), computer vision, system design

Work Experience

AI Intern — Zummit Africa

Sep 2025 - Jul 2026

- Developed and deployed backend systems automating classification and prioritization of legal documents
- Built ML pipelines and REST APIs processing data in real time, reducing manual workload for the team
- Owned backend infrastructure including API design and multi-tenant data isolation
- Integrated AI models into production services, maintaining secure and stable inference under real usage
- Debugged and iterated on model accuracy and backend stability through continuous testing

Lead Frontend Engineer — TLCC.ng

Oct 2025 – Sep 2026

- Owned the frontend platform architecture as the sole engineer responsible for client-side performance
- Built and maintained the full frontend experience for a production platform using HTML, CSS, and JavaScript
- Integrated dynamic content and interactive features with backend APIs
- Managed application state and user workflows, balancing performance with maintainability

Backend Engineering Lead — Techies Collab — CuratED (TCU 7.0)

Apr 2025 – Aug 2025

- Designed and implemented scalable Django REST APIs for content aggregation and progress tracking
- Architected modular backend services and enforced strict API contracts across teams
- Participated in sprint planning, pull request review, and architectural decisions with the team
- Improved test coverage and developer workflows, reducing bugs before deployment

Backend Developer — HealthTech for Africans × Alabama A&M Univ

Sep 2024 – Nov 2024

- Extended backend features for health reporting and resource coordination tools
- Designed RESTful APIs supporting data collection and cross-system integration
- Worked in agile cycles with an international, cross-functional team

PROJECTS

Failures — Deterministic Failure-Mode Guardrails for AI Coding Agents

[failures.pxxl.click](#)

github.com/mayowa-kalejaiye/Failures | Python, MCP, Static Analysis, AST

- Authored a formal model of distributed failure modes across 11 dimensions (atomicity, idempotency, timeout/ambiguous outcome, concurrency, ordering, consistency, availability, resource exhaustion, recovery, observability, retry safety) and implemented it as a 13-tool Model Context Protocol server at zero model-token cost
- Built a deterministic evaluation harness scoring 10 resilience criteria and ran a blind study across 3 commercial AI coding agents over 6 production-style scenarios (17 matched output pairs) with frozen prompts, hashes and manifests
- Published as a citable paper on Zenodo with a reproducible artifact archive; released on PyPI as failures-mcp

GigScout — AI Job Matching SaaS

[gigscout.pxxl.click](#)

- Built and launched an AI job feed end to end in two weeks: a multi-source scanning architecture, an AI scoring engine, BYOK AI integration, job-kit generation, and a Telegram delivery bot — taking it from zero to a live SaaS with paying users
- Reached 37 signups and converted 5% to paid within three weeks of launch, with a first sale on launch day; delivered 740+ AI-scored matches across 19,717 jobs indexed from 8 job boards
- Held total all-time AI spend to \$0.31 while serving those users, by routing inference through the user's own API key rather than absorbing the cost

Tradeoff — AI-Powered Engineering Reasoning Engine

[tradeoff.pxxl.run](#)

- Engineered a multi-LLM (OpenAI, Claude, Gemini) reasoning pipeline to analyze and pressure-test system architectures
- Developed a structured data extraction engine to capture edge cases and dynamically render interactive Mermaid.js diagrams
- Built a full-stack Next.js and React application featuring a dual-rendering system for web interactivity and PDF report generation

EDUCATION

- Software Engineering, MIVA Open University

Expected Dec 2026